American Computer Science League

2022-2023 ● Contest 2: Binary Counting ● Junior Division

PROBLEM: Given a string of characters found on the keyboard, convert each character in the string to the binary equivalent of its ASCII code. In the resulting concatenated string, search for the increasing sequence of binary numbers starting with 0, 1, 10, 11, … until a number cannot be found anywhere in the string.

EXAMPLE: For the string "Roses are red.", convert it to a concatenated string of binary numbers using each character's ASCII code as follows:

Char	ASCII	Binary
R	82	01010010
o	111	01101111
s	115	01110011
e	101	01100101
s	115	01110011
sp	32	00100000
a	97	01100001

Char	ASCII	Binary
r	114	01110010
e	101	01100101
sp	32	00100000
r	114	01110010
e	101	01100101
d	100	01100100
.	46	00101110

Now search for binary numbers beginning with 0 in the following string:

01010010 01101111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 00101110

In this concatenated string (with spaces for readability only), the binary numbers 0, 1, 10, 11, 100, 101, 110, 111, 1000, 1001, 1010, 1011, 1100, 1101, 1110, 1111, and 10000 are found, but not 10001. 10000 is the last one found in the string so the output is its decimal equivalent which is 16.

INPUT: Input a string that will be converted to binary using each character's ASCII code. The string will be fewer than 200 characters.

OUTPUT: Output an integer representing the decimal number equivalent to the last binary number that can be found in the resulting string starting with 0, 1, 10, 11, …

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SAMPLE INPUT:	SAMPLE OUTPUT:
1. Roses are red.	1. 16
2. A is Alpha; B is Bravo; C is Charlie.	2. 20
3. A stitch in time saves nine.	3. 14
4. 1, 2: Buckle my shoe! 3, 4: Shut the door!	4. 30
5. Is HackerRank the platform used by ACSL?	5. 61

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TEST DATA

TEST INPUT:

1. What was the first computer programming language you learned?
2. Lions and Tigers and Bears, Oh My! This is from The Wizard of Oz.
3. Knock, knock. Who is there? Hawaii. Hawaii who? I am good, Hawaii you?
4. How do you use HackerRank to do each ACSL competition?
5. ~{w|x|y|z}

TEST OUTPUT:

1. 61
2. 30
3. 64
4. 33
5. 20

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问题：给定键盘上的字符串，将字符串中每个字符转换为与其 ASCII 码等值的二进制数。生成的连接字符串中，搜索以 0、1、10、11…开始的递增二进制数数列， 直到出现一个在字符串中任意位置都无法被找到的二进制数。

示例：对于字符串“ Roses are red. ”，运用如下所示每个字符的 ASCII 码，将其转换成二进制数的连接字符串。

字符	ASCII	二进制数
R	82	01010010
o	111	01101111
s	115	01110011
e	101	01100101
s	115	01110011
sp	32	00100000
a	97	01100001

字符	ASCII	二进制数
r	114	01110010
e	101	01100101
sp	32	00100000
r	114	01110010
e	101	01100101
d	100	01100100
.	46	00101110

现在搜索下列字符串中从 0 开始的二进制数：

01010010 01101111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 00101110

在这个连接字符串中（空格仅用于阅读），可以找到二进制数 0、1、10、11、100、101、110、111、1000、1001、1010、1011、1100、1101、1110、1111 和 10000，但不能找到10001。10000 是字符串中找到的最后一个二进制数，因此输出与它等值的十进制数，即16。

输入：输入即将使用其中每个字符的 ASCII 码转换为二进制数的字符串。
字符串将少于 200 个字符。

输出：输出一个整数，表示以 0、1、10、11... 开头的字符串中可以找到的最后一个二进制数的十进制等值数。

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样本输入：	**样本输出：**
1. Roses are red.	1. 16
2. A is Alpha; B is Bravo; C is Charlie.	2. 20
3. A stitch in time saves nine.	3. 14
4. 1, 2: Buckle my shoe! 3, 4: Shut the door!	4. 30
5. Is HackerRank the platform used by ACSL?	5. 61

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测试数据

测试输入：

1. What was the first computer programming language you learned?
2. Lions and Tigers and Bears, Oh My! This is from The Wizard of Oz.
3. Knock, knock. Who is there? Hawaii. Hawaii who? I am good, Hawaii you?
4. How do you use HackerRank to do each ACSL competition?
5. ~{w|x|y|z}

测试输出：

1. 61
2. 30
3. 64
4. 33
5. 20